First Light Strategic Memo

First Light — Follow-up for Irina (VP)

To: Irina, VP
From: Jack Al-Kahwati, First Light
Re: Strategic direction, pricing, path to scale
Date: July 2026

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1. Stage (clear answer)

Dimension	Status
Revenue	Pre-revenue; bootstrapped
Compose (software)	Internally dogfooded on all FL-1 and customer-path boards; launching paid enterprise pilots Q3 2026
FL-1 (hardware)	EVT complete; mechanical DFM-ready; founding units at $49,500; reservations open ($2,500 refundable deposit)
Prior capital	No outside capital on First Light; prior companies raised ~$3M

What exists today: A working closed-loop stack — design interview → block-composed KiCad PCB → auto-route/DRC → BOM/fab package → firmware → FL-1 test plan. We use it daily to build our own instrument boards (relay matrix, DUT monitor, cal reference, backplane, IoT/telemetry boards).

What is roadmap (honest): Higher layer count, high-speed interfaces, and full spacecraft avionics classes — not claimed as shipped today.

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2. North Star (one primary motion)

Primary business: Enterprise Compose — AI-native board synthesis for defense, space, and advanced hardware labs.

Wedge: FL-1 autonomous bring-up station — capex replacement for fragmented lab instruments, tightly integrated with Compose test plans.

Attach: Fab routing margin on boards designed and ordered through the platform.

We are **not** pursuing PLG at $49/month for production customers. Free tier remains for evaluation; production is sales-led.

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3. What Compose designs today

Compose turns a natural-language block spec into a placed, routable KiCad board:

- **IoT / telemetry:** MCU + LoRa/GNSS/cellular + sensors (I2C, sourced from DigiKey + datasheets)
- **FL-1 instrument family:** Relay/probe matrix, DUT power monitor, calibration reference chain, instrument bus, board-ID EEPROM
- **Industrial blocks:** CAN transceiver, stepper driver, DC current monitor, motor PWM outputs
- **Outputs per run:** `.kicad_pcb`, BOM, fab package, firmware scaffold, `.devices.json` manifest, FL-1 test plan

Typical designs today: **4-layer**, block-composed embedded and instrument PCBs. Compound specs work (e.g. "IMU + temperature sensor + LoRa tracker").

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4. Pricing (value-based, enterprise)

Platform licenses (annual)

Tier	Price	Buyer	Includes
Pilot	$15,000 (90 days; credits toward Year 1)	Eval / single program	5 board programs, onboarding
Team	$48,000/yr ($4,000/mo)	Startup lab, small prime team	3 seats, 24 board programs/yr, support
Enterprise	$120,000–$150,000/yr	OEM, prime, gov lab	10+ seats, 60+ programs, SLA, SSO
Defense / ITAR	$150,000–$350,000/yr	Space Force, classified programs	Dedicated env, compliance attestation

Not published on website. All production access via sales conversation and private offer.

Board program credits (usage)

Complexity	Examples	Credits	Overage
Simple	MCU + power + 1–2 blocks	1	$2,000
Standard	Multi-block IoT, FL-1 bus, relay matrix	2–3	$5,000
Instrument	DUT monitor, cal ref, sourced ICs	3–5	$8,000
Revision	Re-spin from prior program	0.5	$1,000

Attach revenue

- **Fab routing:** 10% of PCB fab + assembly orders through Compose
- **FL-1 bundle:** $49,500 founding unit includes 12 months Team ($48k value)

What we removed from GTM

- **$49/month Pro** for production use — kept only as optional hobby/student tier if needed; not marketed to enterprise ICP

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5. ICP (initial focus)

Primary: Defense and space electronics labs — 5–50 board spins/year, $80k+ bench instrument capex (characterization labs run $150k–$300k), schedule pressure, ITAR/export sensitivity.

Secondary: Advanced hardware startups (robotics, autonomy, satcom payloads) and OEM innovation labs.

Not initial ICP: Consumer electronics at Apple/Toyota scale (future expansion; dilutes Year 1 focus).

Economic buyer: CFO, VP Ops, program manager — not the EE who uses the tool daily.

Adoption note: Engineers may resist (“NIH”); we sell time-to-market and capex reduction to budget holders, then enable EEs with pilot success.

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6. Alternative cost (ROI anchor)

Per **one board program** (traditional vs First Light) — see attached ROI calculator.

Traditional (in-house, moderate-complexity embedded/instrument board):

- 1 senior EE × 2.5 months loaded @ $20k/mo → **$50,000**
- Test/fixture + contractor layout/firmware NRE → **$15,000**
- 2 fab spins @ $3,000 → **$6,000**

- Bench instruments ($80k capex, 5-yr amortized, spread across annual programs) → **~$1,300**
- **≈ $72,000 per program** (≈ $868k/yr at 12 programs)

First Light (Enterprise, Year 1):

- Enterprise Compose → **$120,000/yr**
- 60 board programs included
- FL-1 (one-time) → **$49,500**
- **≈ $172,500 Year 1** for the full annual run rate

At 12 programs/yr: **~$695k Year 1 savings, ~5× ROI, ~2.4-month payback**. Break-even vs traditional: **~2.5 programs/yr**. Value accelerates with every additional spin. (Defense-grade high-speed/RF programs run $150k–$200k+ each — use that framing only with primes who run that board class.)

Benchmarks behind these assumptions (verifiable, 2025–26):

- Fully loaded senior EE: $190k–$280k/yr (BLS ECEC benefits load ~30% on $150k–$200k base)
- Outsourced full board design program at firm rates: $25k–$60k (design-services published rates, ~250 hrs)
- Professional EDA seats: Altium Designer $3.5k–$7.5k/seat/yr (median org contract ~$18k, Vendr); high-end Cadence/Siemens enterprise seats $20k–$50k+
- Dev-tools enterprise SaaS median ACV: ~$85k (range $50k–$150k) — our Enterprise tier sits on-benchmark

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7. Three business lines (chosen weighting)

Line	Year 1 weight	Rationale
Enterprise Compose SaaS	60%	Highest margin; funds everything; smallest ops footprint
FL-1 hardware	25%	Capex + labor wedge; replaces $40k–$80k of bench instruments plus the engineer-hours to run them; pulls Compose into the lab
Fab attach + services	15%	Scales with customer volume; margin without owning factory

We are **not** open-sourcing Compose or selling FL-1 IP broadly (Bambu Lab model). We may offer **manufacture-as-a-service** for select partners (Tempo-style) while keeping the closed loop proprietary.

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8. Moat / defensibility

1. **Closed-loop data:** Design + fab + bring-up + revision outcomes — not just schematic generation
2. **FL-1 validation corpus:** Test plans tied to composed boards; physical proof, not simulation-only
3. **Block library + sourcing pipeline:** Real parts, verified footprints, firmware manifests
4. **Compliance stack:** ITAR-aware workflows, SOC 2 path, customer IP isolation (no training on customer designs)
5. **Supply chain integration:** Design → quote → order in one flow
6. **Relationship density:** Defense/space logos and grant-funded credibility

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9. Path to $100M (illustrative model)

Year 1–2 (2026–2027): Land logos

Metric	Target
Team customers	10 × $48k = $480k
Enterprise customers	5 × $120k = $600k
FL-1 units	15 × $49.5k = $743k
Fab attach	$200k
Year 2 ARR + hardware	**~$2.0M**

Year 3–4 (2028–2029): Expand seats + defense tier

Metric	Target
Team	25 × $48k = $1.2M
Enterprise	20 × $135k = $2.7M
Defense	8 × $250k = $2.0M
FL-1	40 × $45k = $1.8M
Fab attach	$1.5M
Year 4 run-rate	**~$9.2M**

Year 5–7 (2030–2032): Category leader in defense/space board programs

Metric	Target
Enterprise + Defense ACV	120 accounts × $200k avg = $24M
FL-1 + services	$15M
Fab attach	$8M
Manufacture-as-a-service (select)	$53M
Year 7 run-rate	**~$100M**

Full spreadsheet attached: `compose-financial-model.csv`

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10. Use of $4.5M raise

Allocation	Amount	Milestone
Enterprise GTM (2 AEs + 1 solutions engineer)	$1.2M	15 paying logos in 18 months
FL-1 production (first 25 units, CM tooling)	$1.5M	Ship founding customers
Compose product (block library, compliance, enterprise features)	$1.0M	Defense-tier ready
Grants / compliance / legal (ITAR, SOC 2)	$0.5M	Enterprise procurement unlock
Reserve	$0.3M	

Alternative: Software-first path — 8 Enterprise customers ($960k ACV) + FL-1 preorders could fund first production batch with less dilution. Raise accelerates logo velocity and production scale.

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11. Grants and strategic tailwinds

- US onshore electronics manufacturing and defense innovation priorities align with our stack
- Already submitted 15+ grant applications in 2026 (defense/space track)
- Compose + FL-1 fit SBIR/STTR, DIU-adjacent, and lab modernization narratives
- Grant funding targeted for FL-1 production and compliance, equity-free

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12. Next steps

1. Partner meeting — happy to demo live Compose run (design interview → board → BOM → test plan)
2. Provide 2–3 reference conversations from defense/space pipeline (under NDA as needed)
3. Adjust deck pricing page to sales-led motion (done in parallel)

Thank you for the conversation — the enterprise pricing and buyer framing was exactly the push we needed.

Attachments:

- `compose-roi-calculator.csv` — per-customer ROI model (editable assumptions)
- `compose-financial-model.csv` — 7-year revenue path to $100M

— Jack